

Lecture 02 — Option Pricing Theory

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Study notes

These notes follow Lecture 2 (*Option Pricing Theory*). The logical spine is: Itô's lemma $\rightarrow$ the Black–Scholes PDE through delta-hedging $\rightarrow$ the martingale (risk-neutral) formulation $\rightarrow$ the Feynman–Kac bridge between the two $\rightarrow$ the closed-form European call price. The two pricing philosophies (replication/PDE and change-of-measure/expectation) are made to coincide. Where the slides are thin (put–call parity, the full Greek table) I derive the material myself and flag it.

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1 Itô's lemma

1.1 Statement

Let X_t be a one-dimensional Itô diffusion,

$$dX_t = \mu_t dt + \sigma_t dW_t, \quad X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s, \quad (1)$$

with drift μ_t and diffusion coefficient σ_t , and W_t a standard Brownian motion under the relevant measure. Let $g(t, x)$ be once continuously differentiable in t and twice in x . Then $Y_t := g(t, X_t)$ is again an Itô process and

$$dY_t = \left(\frac{\partial g}{\partial t} + \mu_t \frac{\partial g}{\partial x} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 g}{\partial x^2} \right) dt + \sigma_t \frac{\partial g}{\partial x} dW_t. \quad (2)$$

A companion identity is the *Itô product rule*: for two Itô processes Y_t, Z_t ,

$$d(Y_t Z_t) = Z_t dY_t + Y_t dZ_t + dY_t dZ_t, \quad (3)$$

where the cross term $dY_t dZ_t$ is the quadratic covariation increment and is the feature that distinguishes Itô calculus from ordinary calculus.

1.2 Heuristic derivation: why the second-order term survives

The non-classical term in (2) comes from the quadratic variation of Brownian motion. Over $[t, t + dt]$ the increment satisfies $W_{t+dt} - W_t \sim \mathcal{N}(0, dt)$, so dW_t is of order $\sqrt{dt}$ and $(dW_t)^2$ is of order dt . The formal multiplication table is

$$(dW_t)^2 = dt, \quad dW_t dt = 0, \quad (dt)^2 = 0, \quad (4)$$

where the last two are higher order and vanish faster than dt . Performing an informal Taylor expansion of g to second order,

$$dg = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial t^2} (dt)^2 + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} (dX_t)^2 + \frac{\partial^2 g}{\partial x \partial t} dX_t dt + \dots \quad (5)$$

Keeping only terms of order dt or lower, the $(dt)^2$ and $dX_t dt$ terms drop. The quadratic term reduces using the table:

$$(dX_t)^2 = (\mu_t dt + \sigma_t dW_t)^2 = \mu_t^2 (dt)^2 + 2\mu_t \sigma_t dt dW_t + \sigma_t^2 (dW_t)^2 = \sigma_t^2 dt + o(dt). \quad (6)$$

Substituting $dX_t = \mu_t dt + \sigma_t dW_t$ and $(dX_t)^2 = \sigma_t^2 dt$ and collecting the dt and dW_t pieces yields (2).

Remark 1. The extra $\frac{1}{2} \sigma_t^2 g_{xx}$ drift is purely an artefact of the non-vanishing quadratic variation. It is the same term that, in the PDE derivation below, generates the $\frac{1}{2} \sigma^2 S^2 V_{SS}$ diffusion term, and convexity (sign of g_{xx} , i.e. Gamma) is exactly what it measures.

1.3 Worked example: geometric Brownian motion

Let $dS_t = \mu S_t dt + \sigma S_t dW_t$ and take $g(t, S) = \ln S$, so $g_t = 0$, $g_S = 1/S$, $g_{SS} = -1/S^2$. Then

$$\begin{aligned} d \ln S_t &= \frac{1}{S_t} dS_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right) \sigma^2 S_t^2 dt \\ &= \frac{1}{S_t} (\mu S_t dt + \sigma S_t dW_t) - \frac{\sigma^2}{2} dt = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t. \end{aligned} \quad (7)$$

Integrating from t_0 to t ,

$$\ln S_t = \ln S_{t_0} + \left(\mu - \frac{\sigma^2}{2} \right) (t - t_0) + \sigma (W_t - W_{t_0}). \quad (8)$$

The $-\frac{1}{2}\sigma^2$ correction is the Itô term: $\mathbb{E}[S_t] = S_{t_0} e^{\mu(t-t_0)}$ grows at rate μ , but $\mathbb{E}[\ln S_t]$ grows at the lower rate $\mu - \frac{1}{2}\sigma^2$ because of Jensen's inequality applied to the concave logarithm.

1.4 Resulting distributions

From (8), $X := \ln S_t$ is Gaussian,

$$X = \ln S_t \sim \mathcal{N}(\ln S_{t_0} + (\mu - \frac{1}{2}\sigma^2)(t - t_0), \sigma^2(t - t_0)), \quad (9)$$

with density

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi(t - t_0)}} \exp \left(-\frac{(x - \ln S_{t_0} - (\mu - \frac{1}{2}\sigma^2)(t - t_0))^2}{2\sigma^2(t - t_0)} \right). \quad (10)$$

The price $S_t = e^X$ is therefore *log-normal*: for $y > 0$, $F_S(y) = \mathbb{P}(S_t \leq y) = \mathbb{P}(X \leq \ln y) = \int_{-\infty}^{\ln y} f_X(x) dx$, and the log-normal density (by the change of variables $s = e^x$, with extra factor $1/s$) is

$$f_S(s) = \frac{1}{s \sigma \sqrt{2\pi(t - t_0)}} \exp \left(-\frac{(\ln \frac{s}{S_{t_0}} - (\mu - \frac{1}{2}\sigma^2)(t - t_0))^2}{2\sigma^2(t - t_0)} \right), \quad s > 0. \quad (11)$$

This is the density that the pricing integral in Section 5 integrates the payoff against.

2 The Black–Scholes PDE via delta-hedging

2.1 Set-up

Assume the underlying obeys geometric Brownian motion under the real-world (physical) measure P , and a risk-free bank account M_t :

$$dS_t = \mu S_t dt + \sigma S_t dW_t^P, \quad dM_t = r M_t dt. \quad (12)$$

Let $V(t, S_t)$ denote the option price. Build a *self-financing* hedging portfolio that is long one option and short $\Delta(t, S_t)$ units of the underlying:

$$\Pi(t, S_t) = V(t, S_t) - \Delta(t, S_t) S_t. \quad (13)$$

Self-financing means no capital is injected or withdrawn; over $[t, t + dt]$ the position Δ is held fixed (it is $\mathcal{F}_t$ -measurable, chosen on current information), so

$$d\Pi_t = dV_t - \Delta(t, S_t) dS_t. \quad (14)$$

2.2 Apply Itô to the option value

By Itô's lemma applied to $V(t, S_t)$, with $(dS_t)^2 = \sigma^2 S_t^2 dt$,

$$dV_t = V_t dt + V_S dS_t + \frac{1}{2} V_{SS} (dS_t)^2 = (V_t + \mu S_t V_S + \frac{1}{2} \sigma^2 S_t^2 V_{SS}) dt + \sigma S_t V_S dW_t^P, \quad (15)$$

where subscripts denote partial derivatives ($V_t = \partial V / \partial t$, $V_S = \partial V / \partial S$, $V_{SS} = \partial^2 V / \partial S^2$).

2.3 Eliminate the risk

Substituting dV_t and $dS_t = \mu S_t dt + \sigma S_t dW_t^P$ into (14):

$$d\Pi_t = (V_t + \mu S_t V_S + \frac{1}{2} \sigma^2 S_t^2 V_{SS} - \Delta \mu S_t) dt + (\sigma S_t V_S - \Delta \sigma S_t) dW_t^P. \quad (16)$$

The portfolio is *locally riskless* if and only if the stochastic (dW^P) term vanishes, which forces the hedge ratio

$$\boxed{\Delta(t, S_t) = V_S} \quad (17)$$

Note that the drift μ also disappears from the remaining dt term once $\Delta = V_S$ is substituted: the $\mu S_t V_S$ and $-\Delta \mu S_t = -\mu S_t V_S$ terms cancel. This is the precise reason μ never enters the price. With (17),

$$d\Pi_t = (V_t + \frac{1}{2} \sigma^2 S_t^2 V_{SS}) dt. \quad (18)$$

2.4 Impose no-arbitrage

A locally riskless portfolio must earn the risk-free rate (otherwise an arbitrage exists): $d\Pi_t = r\Pi_t dt$. Using $\Pi_t = V - V_S S_t$,

$$V_t + \frac{1}{2} \sigma^2 S_t^2 V_{SS} = r(V - S_t V_S). \quad (19)$$

Rearranging gives the **Black–Scholes PDE**,

$$\boxed{V_t + r S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS} - r V = 0} \quad (20)$$

a second-order, linear, parabolic PDE, solved backward from the terminal payoff $V(S, T) = g(S)$.

2.5 Modelling assumptions

The derivation rests on: (i) GBM dynamics for S ; (ii) constant, known r and σ ; (iii) no transaction costs in the continuous hedge; (iv) no dividends over the option's life; (v) no arbitrage (and implicitly continuous trading / market completeness). Relaxing any of these breaks either the constant-coefficient PDE or the perfect replication argument.

Remark 2 (The five pricing parameters, and the absent one). *The European call price solves (20) and is a function of S , t , σ , K , T , r . The growth rate μ is conspicuously absent from (20) and hence from the price. Economically: risk preferences are hedged away; only the volatility and the cost of carry r matter. This is the seed of risk-neutral valuation.*

3 Martingales and risk-neutral valuation

3.1 Martingale definition

A process $X = \{X_t : t \geq 0\}$ is a *martingale* with respect to a filtration $\{\mathcal{F}_t\}$ if X_t is adapted, $\mathbb{E}|X_t| < \infty$, and for all $0 \leq s < t$,

$$\mathbb{E}[X_t | \mathcal{F}_s] = X_s. \quad (21)$$

Two examples used repeatedly:

- (a) Brownian motion itself. For $s < t$, $\mathbb{E}[W_t | \mathcal{F}_s] = \mathbb{E}[W_t - W_s | \mathcal{F}_s] + W_s = W_s$, because the increment $W_t - W_s$ is independent of $\mathcal{F}_s$ and has mean zero.
- (b) The *stochastic (Doléans) exponential* $X_t = \exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$. Using the Gaussian exponential moment $\mathbb{E}[e^{\sigma Z - \frac{1}{2}\sigma^2}] = 1$ for $Z \sim \mathcal{N}(0, 1)$,

$$\mathbb{E}[X_t | \mathcal{F}_s] = e^{\sigma W_s - \frac{1}{2}\sigma^2 s} \mathbb{E}\left[e^{\sigma(W_t - W_s) - \frac{1}{2}\sigma^2(t-s)}\right] = X_s, \quad (22)$$

since $W_t - W_s \sim \mathcal{N}(0, t - s)$ makes the conditional expectation equal to one.

- (c) Any Itô integral $X_t = \int_0^t g(s) dW_s$ with $\mathbb{E} \int_0^T g^2 ds < \infty$ is a martingale. This is the workhorse fact: *a drift-free Itô process (one with no dt term) is a martingale.*

3.2 Risk-neutral (equivalent martingale) measure

A *risk-neutral measure* Q is a probability measure equivalent to P under which the *discounted* asset price $e^{-rt}S_t$ is a martingale:

$$S_t = \mathbb{E}^Q\left[e^{-r(T-t)}S_T | \mathcal{F}_t\right]. \quad (23)$$

Suppose under Q the asset still follows a GBM but with unknown drift μ' , $dS_t = \mu'S_t dt + \sigma S_t dW_t^Q$. We determine μ' by demanding the martingale property. By the product rule (3), with $d(e^{-rt}) = -re^{-rt} dt$ and the cross term vanishing (the bank account is finite variation),

$$\begin{aligned} d(e^{-rt}S_t) &= S_t d(e^{-rt}) + e^{-rt} dS_t + d(e^{-rt}) dS_t \\ &= -re^{-rt}S_t dt + e^{-rt}(\mu'S_t dt + \sigma S_t dW_t^Q) = (\mu' - r)e^{-rt}S_t dt + e^{-rt}\sigma S_t dW_t^Q. \end{aligned} \quad (24)$$

Integrating,

$$e^{-rt}S_t = S_0 + \int_0^t (\mu' - r)e^{-rv}S_v dv + \int_0^t e^{-rv}\sigma S_v dW_v^Q. \quad (25)$$

The stochastic integral is a martingale; the Lebesgue integral (the dv drift) is not, unless it vanishes. Hence $e^{-rt}S_t$ is a martingale *iff* $\mu' - r = 0$. Therefore under Q ,

$$\boxed{dS_t = rS_t dt + \sigma S_t dW_t^Q} \quad (26)$$

The change of measure shifts the drift from μ to r while leaving σ untouched (Girsanov: only the drift is affected). Equivalence means P and Q agree on which events are impossible ($Q(A) = 0 \Leftrightarrow P(A) = 0$), so σ — a path/quadratic-variation property — is preserved.

3.3 Discounted option price is a Q -martingale

By the same no-arbitrage logic, the discounted *option* price $e^{-rt}V(t, S_t)$ is a Q -martingale. Using the product rule and Itô under the Q -dynamics (26),

$$dV = (V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}) dt + \sigma SV_S dW_t^Q, \quad (27)$$

and therefore

$$d(e^{-rt}V) = e^{-rt} (V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV) dt + e^{-rt}\sigma SV_S dW_t^Q. \quad (28)$$

For $e^{-rt}V$ to be a martingale the drift must vanish, which is again exactly the Black–Scholes PDE (20). So the martingale route reproduces the PDE route. Conversely, applying the martingale property between t and T gives the **risk-neutral valuation formula**

$$\boxed{V(t, S_t) = \mathbb{E}^Q\left[e^{-r(T-t)}g(S_T) | \mathcal{F}_t\right]} \quad (29)$$

4 Feynman–Kac: the bridge

4.1 Statement

The Feynman–Kac theorem links a terminal-value parabolic PDE to a discounted expectation. Consider

$$\frac{\partial V}{\partial t} + b(x, t) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2(x, t) \frac{\partial^2 V}{\partial x^2} - r(x, t) V = 0, \quad V(x, T) = g(x). \quad (30)$$

Let X_s solve $dX_s = b(X_s, s) ds + \sigma(X_s, s) dW_s$, $s \geq t$, with $X_t = x$. Then

$$V(x, t) = \mathbb{E} \left[e^{-\int_t^T r(X_u, u) du} g(X_T) \mid X_t = x \right]. \quad (31)$$

4.2 Proof sketch

Define the discounted process $Y_s = e^{-\int_t^s r(X_u, u) du} V(X_s, s)$ for $s \geq t$. Applying Itô's lemma (the discount factor contributes $-rV$),

$$\begin{aligned} dY_s &= e^{-\int_t^s r du} \left(V_t + b(X_s, s) V_x + \frac{1}{2} \sigma^2(X_s, s) V_{xx} - r(X_s, s) V \right) ds \\ &\quad + e^{-\int_t^s r du} \sigma(X_s, s) V_x dW_s. \end{aligned} \quad (32)$$

By the PDE (30) the entire ds drift vanishes, leaving Y_s a (local) martingale. Taking expectations of $Y_T - Y_t$ and using $Y_t = V(x, t)$, $Y_T = e^{-\int_t^T r du} g(X_T)$ gives (31). So *solving the PDE* and *computing the discounted expectation* are the same operation, viewed from two sides.

4.3 Worked example

Solve $V_t + \frac{1}{2} \sigma^2 V_{xx} = 0$ with $V(x, T) = x^2$, σ constant. Here $b \equiv 0$, $r \equiv 0$, so $dX_s = \sigma dW_s$, $X_t = x$, giving $X_T = x + \sigma(W_T - W_t) \sim \mathcal{N}(x, \sigma^2(T - t))$. By Feynman–Kac, $V(x, t) = \mathbb{E}[X_T^2]$, and using $\mathbb{E}[X_T^2] = \text{Var}(X_T) + (\mathbb{E}X_T)^2$,

$$V(x, t) = \sigma^2(T - t) + x^2. \quad (33)$$

One checks directly that $V_t = -\sigma^2$ and $\frac{1}{2} \sigma^2 V_{xx} = \frac{1}{2} \sigma^2 \cdot 2 = \sigma^2$ sum to zero, and $V(x, T) = x^2$.

4.4 Applied to Black–Scholes

For the Black–Scholes PDE (20) with $V(S, T) = g(S)$, Feynman–Kac (under the Q -dynamics (26), where $b = rS$ and the discount rate is the constant r) gives

$$V(S_t, t) = e^{-r(T-t)} \mathbb{E}^Q[g(S_T) \mid \mathcal{F}_t], \quad (34)$$

equivalently $e^{-rt} V(S_t, t) = \mathbb{E}^Q[e^{-rT} g(S_T) \mid \mathcal{F}_t]$, i.e. the discounted price is a Q -martingale with the bank account $B_t = e^{rt}$ as *numeraire*.

4.5 Equivalent martingale measure (EMM) and numeraire

More generally, Q is an EMM relative to numeraire B_t if it is equivalent to P and $S_t/B_t = \mathbb{E}^Q[S_T/B_T \mid \mathcal{F}_t]$. Prices are then

$$V_t = B_t \mathbb{E}^Q \left[\frac{g(S_T)}{B_T} \mid \mathcal{F}_t \right]. \quad (35)$$

This abstraction matters later in the course: changing the numeraire (e.g. to the stock or to a bond) changes the convenient measure but not the price.

Remark 3 (Two roads, same destination). *The replication/PDE road chooses $\Delta = V_S$ to kill local risk and invokes no-arbitrage to fix the return. The change-of-measure road posits an EMM Q under which discounted prices are martingales. Feynman–Kac shows the two are mathematically identical: $PDE \text{ drift} = 0 \Leftrightarrow \text{discounted price is a } Q\text{-martingale}$. Computationally this hands us several tools: solve the PDE (analytic / finite differences), Monte Carlo the expectation (29), or—when the risk-neutral density or characteristic function is known—numerically integrate $V = e^{-r(T-t)} \int g(s) f_{S_T|S_t}(s) ds$. The last is the gateway to the Fourier/COS methods that dominate the rest of the course.*

5 Closed-form European call: solving the PDE by expectation

5.1 Risk-neutral terminal law

Under Q , by Itô on $\ln S_t$ (identical to the GBM example but with $\mu \rightarrow r$),

$$S_T = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T^Q\right), \quad S_T = S_0 e^Z, \quad Z \sim \mathcal{N}\left(\left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right). \quad (36)$$

The time-0 European call price with strike K , maturity T , payoff $(S_T - K)^+$ is

$$C_0 = e^{-rT} \mathbb{E}^Q[(S_T - K)^+] = e^{-rT} \int_{\ln(K/S_0)}^{\infty} (S_0 e^z - K) f_Z(z) dz, \quad (37)$$

the lower limit $z = \ln(K/S_0)$ being where $S_0 e^z = K$, i.e. where the call enters the money. Split into two integrals, $C_0 = e^{-rT}(I_1 - I_2)$ with $I_1 = \int S_0 e^z f_Z dz$ and $I_2 = \int K f_Z dz$.

5.2 The $K N(d_2)$ term

The second (easy) integral is just K times a tail probability:

$$I_2 = \int_{\ln(K/S_0)}^{\infty} K f_Z(z) dz = K \mathbb{P}\left(Z \geq \ln \frac{K}{S_0}\right). \quad (38)$$

Standardize $Z = (r - \frac{\sigma^2}{2})T + \sigma\sqrt{T}\xi$ with $\xi \sim \mathcal{N}(0, 1)$:

$$I_2 = K \mathbb{P}\left(\xi \geq \frac{\ln(K/S_0) - (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right) = K \mathbb{P}(\xi \leq d_2) = K N(d_2), \quad (39)$$

using symmetry of the standard normal $\mathbb{P}(\xi \geq -a) = \mathbb{P}(\xi \leq a)$, where

$$d_2 = \frac{\ln(S_0/K) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}. \quad (40)$$

5.3 The $S_0 e^{rT} N(d_1)$ term

For I_1 insert the Gaussian density of Z and complete the square in the exponent. With $m = (r - \frac{\sigma^2}{2})T$ and $v^2 = \sigma^2 T$,

$$S_0 e^z f_Z(z) = \frac{S_0}{\sqrt{2\pi v^2}} \exp\left(z - \frac{(z - m)^2}{2v^2}\right). \quad (41)$$

Completing the square, $z - \frac{(z-m)^2}{2v^2} = -\frac{(z-(m+v^2))^2}{2v^2} + m + \frac{1}{2}v^2$, and $m + \frac{1}{2}v^2 = rT$. Hence

$$S_0 e^z f_Z(z) = S_0 e^{rT} \tilde{f}(z), \quad (42)$$

where $\tilde{f}$ is the $\mathcal{N}(m+v^2, v^2)$ density (the original mean shifted up by v^2 — this is the Girsanov/measure-change-to-stock-numeraire shift). Therefore

$$I_1 = S_0 e^{rT} \mathbb{P}\left(\tilde{Z} \geq \ln \frac{K}{S_0}\right) = S_0 e^{rT} \mathbb{P}(\xi \leq d_1) = S_0 e^{rT} N(d_1), \quad (43)$$

where, standardizing the shifted variable, the only change versus d_2 is that the mean now carries $+\frac{\sigma^2}{2}$ instead of $-\frac{\sigma^2}{2}$:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = d_2 + \sigma\sqrt{T}. \quad (44)$$

5.4 The Black–Scholes formula

Combining $C_0 = e^{-rT}(I_1 - I_2)$,

$$\boxed{C_0 = S_0 N(d_1) - K e^{-rT} N(d_2), \quad d_{1,2} = \frac{\ln(S_0/K) + (r \pm \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}} \quad (45)$$

The two terms have a clean reading: $S_0 N(d_1)$ is the (discounted, in numeraire terms) value of receiving the stock conditional on exercise, and $K e^{-rT} N(d_2)$ is the present value of paying the strike, weighted by the risk-neutral exercise probability $N(d_2) = \mathbb{Q}(S_T > K)$.

5.5 European put and put–call parity

The slides obtain the put only by quoting put–call parity; the parity itself is not derived there. The derivation below is mine. Consider the static portfolio: long one call, short one put, both struck at K with maturity T . Its terminal payoff is

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K \quad (46)$$

for every S_T (check the two regimes $S_T \geq K$). A payoff $S_T - K$ at time T is replicated today by holding one unit of stock and shorting K zero-coupon bonds, with present value $S_0 - K e^{-rT}$. By no-arbitrage the prices must match:

$$\boxed{C_0 - P_0 = S_0 - K e^{-rT}} \quad (47)$$

Hence $P_0 = C_0 - S_0 + K e^{-rT}$. Substituting (45) and using $N(-x) = 1 - N(x)$,

$$P_0 = K e^{-rT} N(-d_2) - S_0 N(-d_1), \quad (48)$$

the Black–Scholes put formula. Parity is model-free: it uses only no-arbitrage and a constant rate, not the GBM assumption.

6 The Greeks

The slides name only Delta and Gamma explicitly and mention Theta in passing; the closed forms below for the call I supply for completeness. Greeks are sensitivities of the price to inputs and are the hedging coefficients.

Delta $\Delta = \frac{\partial V}{\partial S}$ is the number of units of underlying needed to hedge — exactly the $\Delta = V_S$ of (17). For the call, $\Delta_{\text{call}} = N(d_1) \in (0, 1)$.

Gamma $\Gamma = \frac{\partial^2 V}{\partial S^2}$ measures how fast Delta moves. Small $\Gamma \Rightarrow$ Delta is stable and one can re-hedge infrequently; large $\Gamma \Rightarrow$ frequent rebalancing. For the call, $\Gamma = \frac{N'(d_1)}{S_0 \sigma \sqrt{T}} > 0$, with N' the standard normal density. Gamma is the discrete cost of the continuous-hedging idealization.

Theta $\Theta = \partial V / \partial t$, the time decay (the slides call it sensitivity to time to maturity). Note that (20) can be read as an algebraic relation among the Greeks: $\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rV$, so the Greeks are not independent — the PDE ties them together. Vega ($\partial V / \partial \sigma$) and Rho ($\partial V / \partial r$) complete the usual set but lie outside the lecture's scope.

7 Coverage check and likely exam targets

Covered by the slides and derived here: Itô's lemma and the order-counting heuristic; GBM solution and (log-)normal densities; the delta-hedging derivation of the BS PDE; the martingale / risk-neutral derivation and why $\mu \rightarrow r$; Feynman–Kac with proof and the BS specialization; EMM / numeraire; the closed-form call via the log-normal expectation; put–call parity (derived here, only quoted on the slides).

Thin on the slides / supplied here: put–call parity proof; the put formula; closed-form Greeks beyond the verbal Delta/Gamma. These are standard and exam-likely, so they are included rather than left blank.

Most probable oral questions. (1) “Where does the $\frac{1}{2}\sigma^2 S^2 V_{SS}$ term come from?” — the quadratic variation $(dW)^2 = dt$. (2) “Why does μ disappear?” — the μSV_S terms cancel once $\Delta = V_S$; equivalently the drift is replaced by r under Q . (3) “Show the PDE and the expectation give the same price” — Feynman–Kac, drift = 0 $\Leftrightarrow$ discounted price is a Q -martingale. (4) “Interpret $N(d_2)$ ” — the risk-neutral probability of finishing in the money, $\mathbb{Q}(S_T > K)$. (5) “Derive parity” — the static $S_T - K$ replication argument.